Algebra 1
Unit 13
Lesson 1 Practice

Name____________________
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The Yang family is adding a second floor to their home. They were approved by their bank for a loan of $\$100{,}000$ with a simple interest rate of 4.85%.

1. Write an equation that models the Yang's loan.

2. Rewrite the equation in slope-intercept form.

3. Complete the table.

Key Feature	**Value**	**Interpretation in Context**
y-intercept		
slope		

4. What is the total the Yang's will have to pay if they pay off the loan in 10 years?

5. How many years did it take the Yang's to pay off the loan if the total they paid was $\$135{,}000$?

Mekhi just graduated college and is starting his first job. He is investing money into a savings account to plan for retirement. He has $\$15{,}000$ to invest in a savings account with a 3.25% APR.

6. Write an equation that models Mekhi's savings account.

7. Write the equation in slope-intercept form.

8. Complete the table.

Key Feature	**Value**	**Interpretation in Context**
y-intercept		
slope		

9. How much money will be in the savings account if Mekhi wants to retire 30 years later?

10. How many years would Mekhi have to work to retire with $\$35{,}000$?

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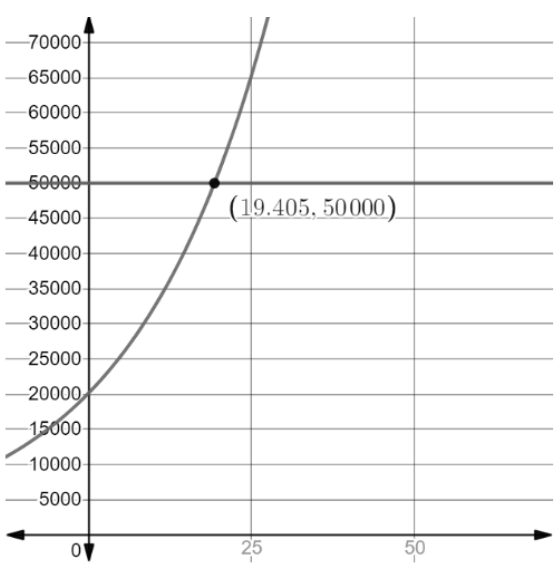
Juan deposited $\$20{,}000$ in a savings account with a 4.75% APR compounded quarterly.

1. Write an equation that models the amount of money Juan has in the account.

2. What is the balance in the account after 8 years?

3. What is the balance in the account after 12 years?

4. Use the graph to determine when the savings account will reach $\$50{,}000$.



For questions 5-10, complete the table below.

	Simple Interest	Compound Interest
Investment amount and APR	$\$150{,}000$ is invested at an interest rate of 5.25% APR	$\$150{,}000$ is invested at an interest rate of 5.25% APR compounded monthly
Write the equation.	5. $A = P(1 + rt)$	6. $A = P\left(1 + \frac{r}{n}\right)^{nt}$
Determine the amount after 5 years.	7.	8.
Determine the amount after 25 years.	9.	10.

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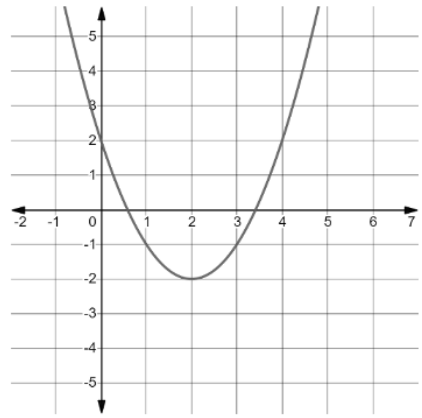
The table shows three different functions.

<table>
<tr><th>Function A</th><th>Function B</th><th>Function C</th></tr>
<tr><td>

x	y
-2	3
-1	6
0	9
1	12
2	15

</td><td>
</td><td>$y = 5(1.4)^x$</td></tr>
</table>

1. Determine which type of function that each represents.

Function A	Function B	Function C
○ Exponential ○ Linear ○ Quadratic	○ Exponential ○ Linear ○ Quadratic	○ Exponential ○ Linear ○ Quadratic

2. Which function has the largest y-intercept?

3. Compare the ranges of the functions and indicate any similarities and differences.

4. Compare the domains of the functions and indicate any similarities and differences.

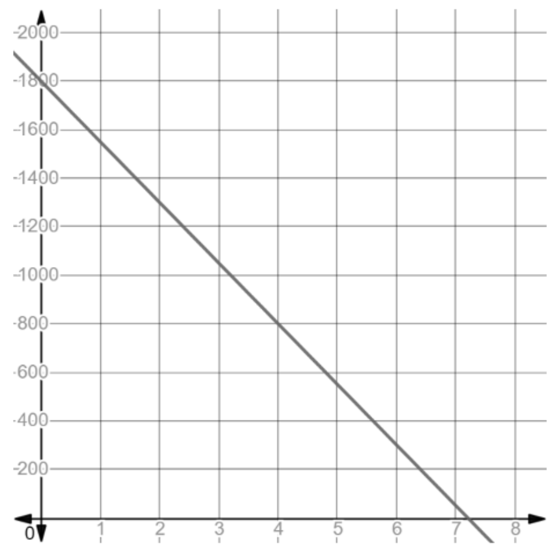
5. As $x \to \infty$, which function has larger values of y?

The table shows a table of values representing the revenue, in billions of dollars, from sales of digital movies, where $x = 0$ is 2010, and a graph representing the revenue, in billions of dollars, from sales of digital books, where $x = 0$ is 2010.

Digital Movie Sales (billions of $)

x	y
0	2057.35
1	1833.96
2	1634.83
3	1457.32
4	1299.08

Digital Book Sales (billions of $)



6. Approximate the x-intercept for the function that represents the sale of digital books.

7. Interpret the x-intercept of the function that represents the sale of digital books.

8. What is the domain of the function representing the sale of digital books?

9. Which function has a larger y-intercept?

10. What type of function best describes the table of values that represents digital movie sales? Explain your choice.

- Exponential
- Linear
- Quadratic

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Determine which family of functions each transformation is part of.

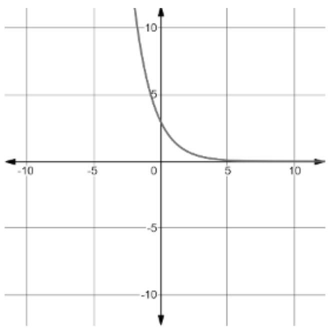
Representation of Transformation	**Equation of Parent Function**	

1.

x	y
-2	-9
-1	-11
0	-13
1	-15
2	-17

- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$
- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1^x}{2}$

2.



- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$
- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1^x}{2}$

3.

x	y
-2	-12
-1	-3
0	0
1	-3
2	-12

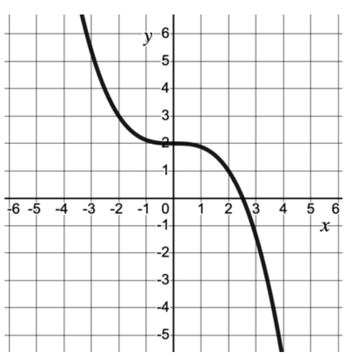
- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$
- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1^x}{2}$

4.

$$y = \frac{2}{3}x + 6$$

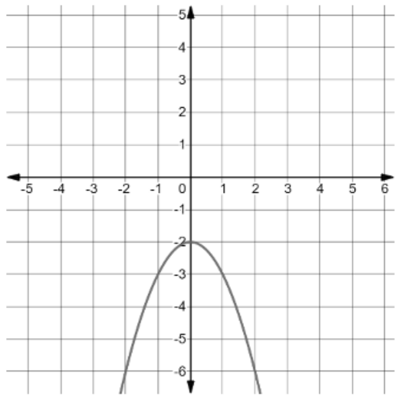
- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$
- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1^x}{2}$

5.



- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$
- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1^x}{2}$

6.



- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$

- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1}{2}^{x}$

7.

x	-2	-1	0	1	2
y	18	6	2	$\frac{2}{3}$	$\frac{2}{9}$

- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$

- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1}{2}^{x}$

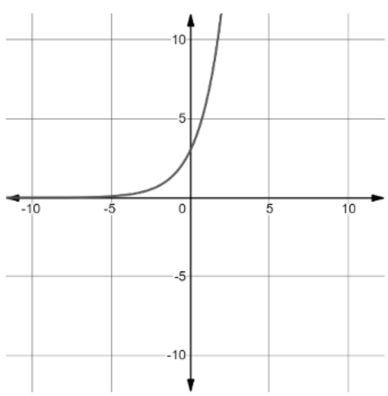
8.

$$y = \sqrt{x + 6} - 4$$

- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$

- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1}{2}^{x}$

9.



- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$

- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1}{2}^{x}$

10.

$$y = 2|x - 5|$$

- ○ $y = x$
- ○ $y = x^2$
- ○ $y = x^3$
- ○ $y = \sqrt{x}$

- ○ $y = |x|$
- ○ $y = 2^x$
- ○ $y = \frac{1}{2}^{x}$

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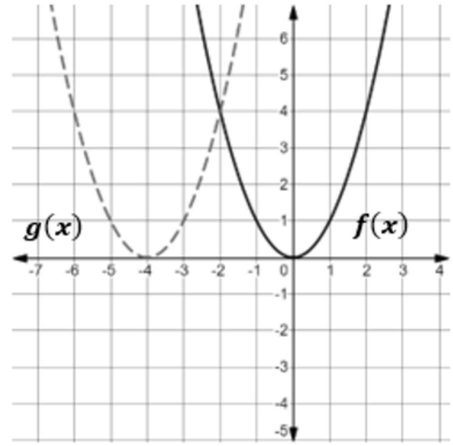
The graphs of $y = f(x)$ and $y = g(x)$ are shown.



1. Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.

2. Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1		-3	
0		-4	
1		-5	

3. Explain the change in the x-values from $f(x)$ to $g(x)$.

4. Complete the statement:

The graph of $g(x)$ is a
- ○ horizontal
- ○ vertical

shift of $f(x)$.

5. Complete the statement:

The shift is seen in the table by
- ○ adding
- ○ subtracting

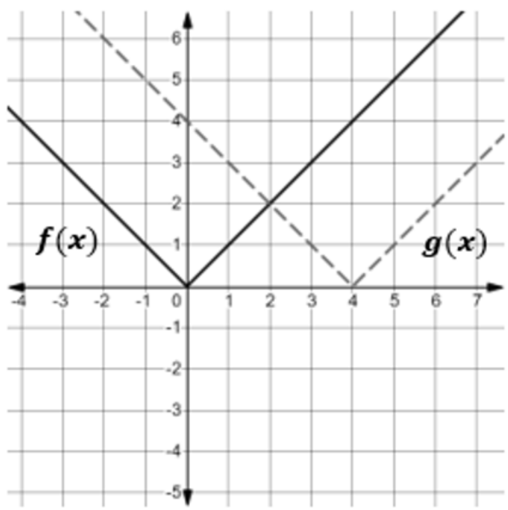
4 to each
- ○ x-value
- ○ y-value

for each corresponding
- ○ x-value.
- ○ y-value.

Complete the table for questions 6-8. The first row has been completed.

Original Equation	**Original Function**	**Shift**	**Transformed Function**
$y = x^2$	$f(x) = x^2$	Right 2	$f(x-2) = (x-2)^2$
6. $y = \|x\|$		Right 4	
7. $y = x^3$		Left 4	
8. $y = \sqrt{x}$		Right 5	

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



9. How does the graph of $f(x)$ compare to the graph of $g(x)$?

10. Two function descriptions are shown. Which function description matches the transformation on the graph? Justify your choice.

- ○ $g(x) = f(x+4)$
- ○ $g(x) = f(x-4)$

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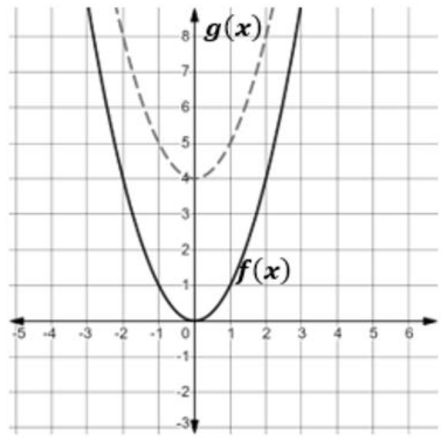
The graphs of $y = f(x)$ and $y = g(x)$ are shown.



1. Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.

2. Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1		-1	
0		0	
1		1	

3. Explain the change in the y-values from $f(x)$ to $g(x)$.

4. Complete the statement:

The graph of $g(x)$ is a ○ horizontal ○ vertical shift of $f(x)$.

5. Complete the statement:

The shift is seen in the table by ○ adding ○ subtracting 4 to each ○ x-value ○ y-value for each corresponding ○ x-value. ○ y-value.

For 6-7, complete the statements to describe how the graph of $g(x) = f(x) - 5$ compares to the graph of $f(x) = |x + 2|$.

6. The graph of $g(x)$ is a

- ○ horizontal
- ○ vertical

shift of $f(x)$.

7. The graph of $g(x)$ is shifted _____ units

- ○ up from
- ○ down from
- ○ to the right of
- ○ to the left of

the graph of $f(x)$.

8. Describe the effect on the x-values of $K(x) = |x - 3|$ for the transformation $J(x) = K(x) + 6$.

9. Describe the effect on the y-values of $v(x) = \frac{3}{4}x + 7$ for the transformation $w(x) = v(x) - 3$.

10. Describe how the graph of $T(x) = S(x - 10)$ compares to the graph of $S(x) = x^2 + 2$.

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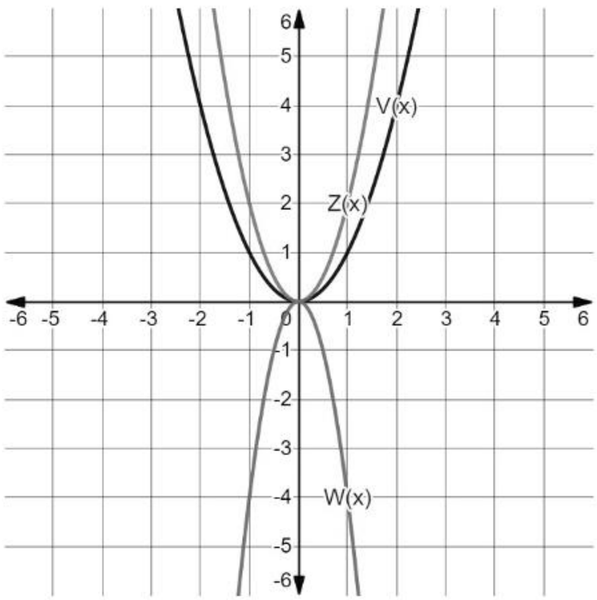
The functions $V(x)$, $W(x)$, and $Z(x)$ are shown on the graph, where $V(x) = x^2$, $W(x) = -4x^2$, and $Z(x) = 2x^2$. The table of values for each function are also shown.



$y = V(x)$		$y = W(x)$		$y = Z(x)$	
x	y	x	y	x	y
-3	9	-3	-36	-3	18
-2	4	-2	-16	-2	8
-1	1	-1	-4	-1	2
0	0	0	0	0	0
1	1	1	-4	1	2
2	4	2	-16	2	8
3	9	3	-36	3	18

Complete the statements below.

1. The coefficient of $V(x) = x^2$ is______.
2. The coefficient of $W(x) = -4x^2$ is______.
3. The y-values of $W(x)$ can be found by multiplying the y-values of $V(x)$ by_____.
4. The coefficient of $Z(x) = 2x^2$ is______.
5. The y-values of $Z(x)$ can be found by multiplying the y-values of $V(x)$ by_____.

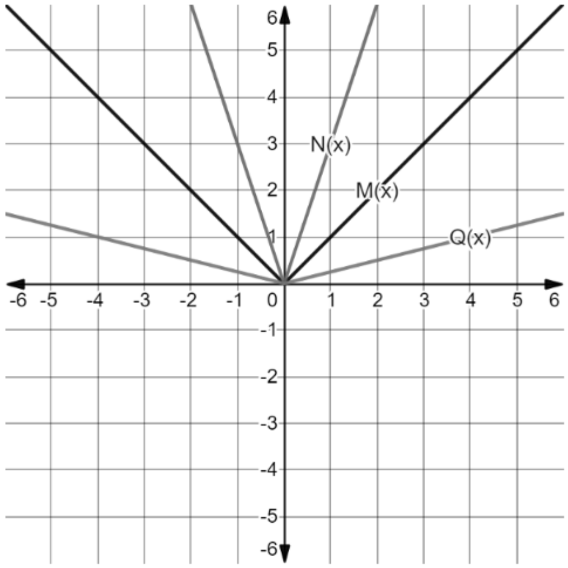
The table of values of the function $y = B(x)$ is shown.

x	-6	-4	-2	0	2	4	6
y	4	7	10	13	16	19	22

6. Complete the sentence to describe the effect on the x-values for the function $y = B(2x)$. The x-values for $B(x)$ will be
   - ○ multiplied
   - ○ divided

   by 2.

The functions $M(x)$, $N(x)$, and $Q(x)$ are shown on the graph, where $M(x) = |x|$, $N(x) = |3x|$, and $Q(x) = |0.25x|$. The table of values for each function are also shown.



$y = M(x)$		$y = N(x)$		$y = Q(x)$	
x	y	x	y	x	y
-2	2	$-\frac{2}{3}$	2	-8	2
-1	1	$-\frac{1}{3}$	1	-1	4
0	0	0	0	0	0
1	1	$\frac{1}{3}$	1	1	4
2	2	$\frac{2}{3}$	2	2	8

Complete the statements below.

7. The x-values of $N(x)$ can be found by
   - multiplying
   - dividing

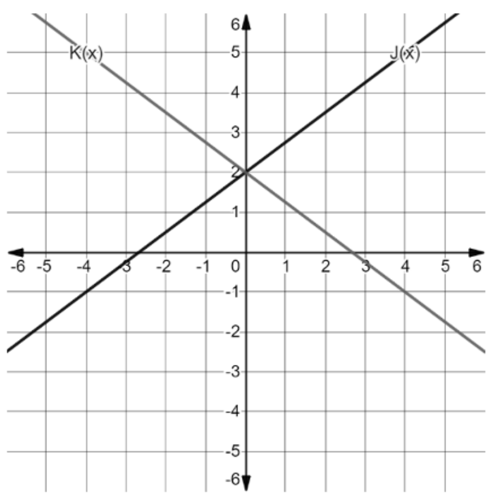
   the x-values of $M(x)$ by_____.

8. The x-values of $Q(x)$ can be found by
   - multiplying
   - dividing

   the x-values of $M(x)$ by_____.

The functions $J(x)$ and $K(x)$ are shown on the graph where $K(x) = J(-x)$. The table of values for each function are also shown.

$y = J(x)$		$y = K(x)$	
x	y	x	y
-2	0.5	-2	3.5
-1	1.25	-1	2.75
0	2	0	2
1	2.75	1	1.25
2	3.5	2	0.5



9. What is the effect on the table of values when the x in $J(x)$ was replaced with $-x$ to create $K(x)$?

10. What is the effect on the graph when the x in $J(x)$ was replaced with $-x$ to create $K(x)$?

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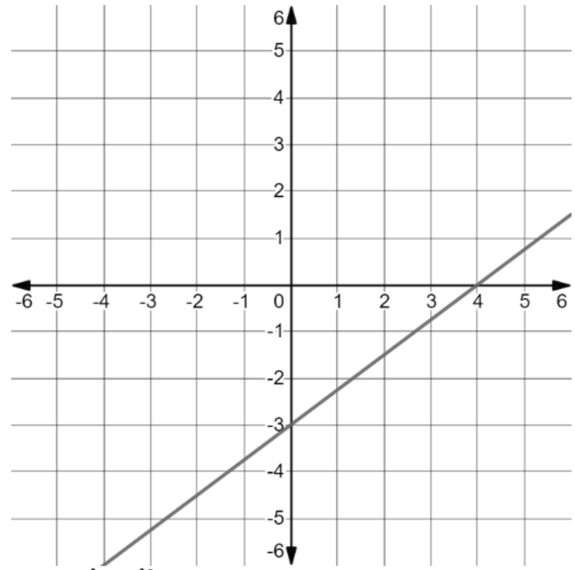
The graph of the function $G(x)$ is shown.



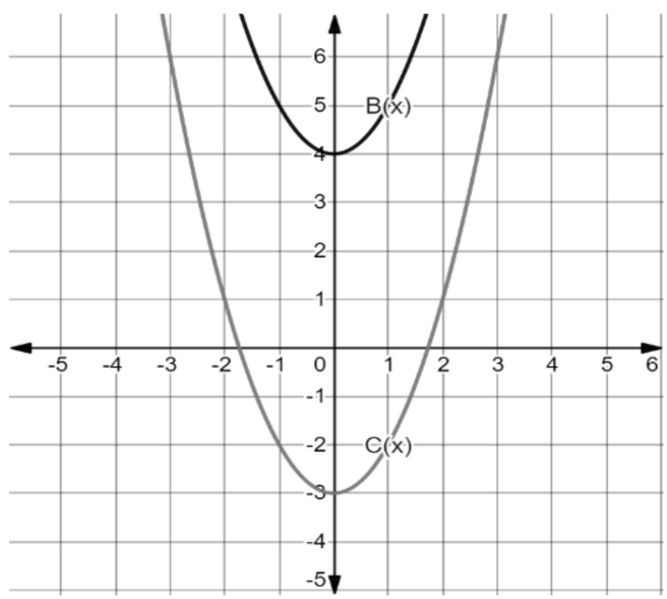
1. Write the function, $G(x)$, that represents the line.

2. Draw a line that is parallel to $G(x)$ with a y-intercept of 3. Label this line $H(x)$.

3. Describe how $G(x)$ can be shifted to coincide with $H(x)$.

4. Write the function $H(X)$ in terms of $G(x)$.

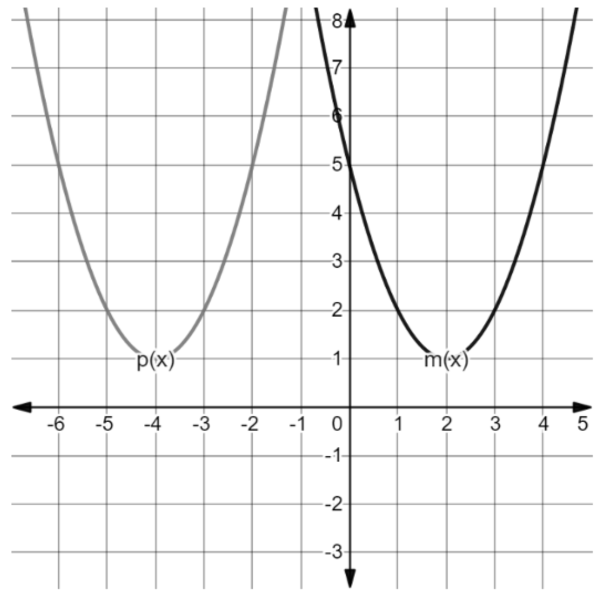
5. The graphs of $B(x)$ and $C(x)$ are snown on the graph.



Write the transformation by completing the equation.

$$C(x) = B(x) + \_\_\_\_$$

The functions $m(x)$ and $p(x)$ are shown on the graph.

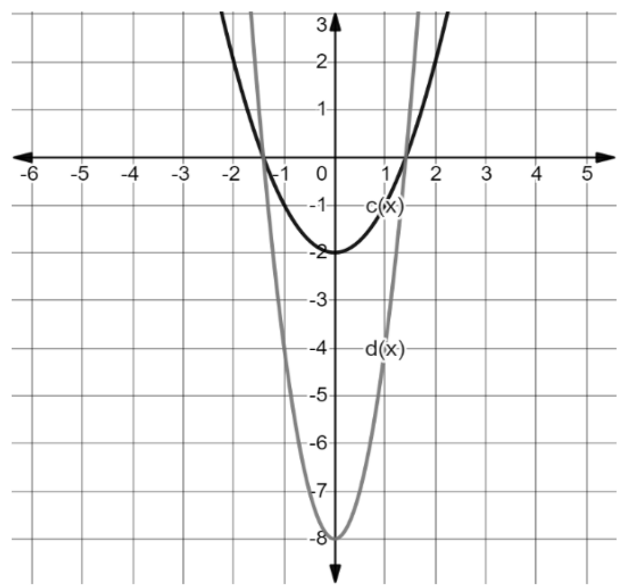


6. Write $m(x)$ in vertex form.

7. Write $p(x)$ in vertex form.

8. Describe how $m(x)$ was transformed to create $p(x)$.

9. The functions $c(x)$ and $d(x)$ are shown on the graph, where $d(x)$ is a transformation of $c(x)$. What is the value of k such that $d(x) = kc(x)$?



10. The table of values for $d(x)$ and $f(x)$ are shown, where $f(x)$ is a transformation of $d(x)$. Write a description of the transformation.

$y = d(x)$		$y = f(x)$	
x	y	x	y
-2	16	-2	20
-1	4	-1	8
0	0	0	4
1	4	1	8
2	16	2	20

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Calista is running a marathon. Based on the temperature on race day and her average marathon time, she calculated that she needs to consume 120 fluid ounces throughout the race. She wants to use both sports drinks and plain water, which she will grab from aid stations along the course. She can get sports drink in 6-ounce cups and water in 8-ounce cups.

1. $6s + 8w$ is a(n)
   - ○ function.
   - ○ equation.
   - ○ expression.

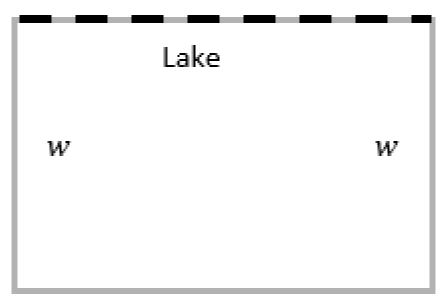
2. The values 6 and 8 are
   - ○ coefficients.
   - ○ terms.
   - ○ constants.

3. The monomials $6s$ and $8w$ are
   - ○ coefficients.
   - ○ terms.
   - ○ constants.

4. $6s + 8w = 120$ is a(n)
   - ○ function.
   - ○ equation.
   - ○ expression.

5. Determine the amount of water Calista needs if she consumes 8 cups of sports drink.

A farmer has $1{,}000$ feet of fencing to enclose his property, which borders a lake. The diagram shows the width, w, of the property and the lake that his property borders. The fencing is not used along the lake.



Explain what each quantity represents in this context.

6. $1000 - 2w$

7. $w(1000 - 2w)$

Midori's parents started a college savings fund when she was born in 2010. The amount of money in the savings fund since 2010 can be represented by the equation $y = 10{,}000(1 + 0.085)^x$, where y is the amount in the college fund and x is the number of years since 2010.

8. The equation $y = 10{,}000(1 + 0.085)^x$ represents
   - ○ exponential growth.
   - ○ exponential decay.

9. The rate of
   - ○ decay
   - ○ growth

   is ____%.

10. What does the coefficient $10{,}000$ represent?

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To make a 14% acid solution, a chemist mixes 4 parts of 2% acid solution with an 18% acid solution. The expression $4(0.02x) + 0.18y$ represents the mixture, where x is the number of ounces of 2% solution and y is the number of ounces of the 18% solution.

Explain what each quantity means.

1. $(0.02x)$

2. $4(0.02x)$

3. $0.18y$

Since 2018, the number of independently contracted food delivery drivers can be modeled by the equation $y = 1{,}675{,}000(1.064)^x$.

4. What does the x variable represent?

5. What does the y variable represent?

6. What does the quantity $1{,}675{,}000$ represent?

7. What is the constant percent rate of change?

The equation $y = -4(x-2)^2 + 20$ models the height, y, in feet, of a ball, x seconds after it is thrown into the air.

Explain what each quantity represents.

8. $x - 2$

9. 20

Michael is buying coffee for $\$5.75$ per bag. He has a coupon for three free bags of coffee. He pays $\$25.50$ for a birthday cake. The expression $5.75(x-3) + 25.50$ can be used to represent the overall cost.

10. What does the quantity $5.75(x-3)$ represent?

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Given a written description, identify the function as linear, quadratic, or exponential. Explain your choice.

1. An investment grows at a constant percent rate of 3.5% APR.

2. The Americans with Disabilities Act (ADA) requires wheelchair ramps to have a 1-inch rise for every 12 horizontal inches of ramp.

3. A ball is thrown from a height of 4 feet. It reaches a maximum height of 35 feet 3 seconds later.

4. A rare species of bird experiences a population decline at a constant percent rate of 3%.

Given a table of values, identify the function as linear, quadratic, or exponential. Explain your choice.

5.

x	y
1	34
2	46
3	50
4	46
5	34

6.

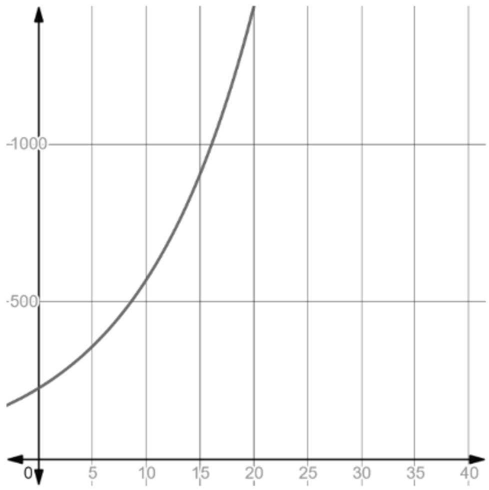
x	y
0	70
1	75
2	80
3	85
4	90

7.

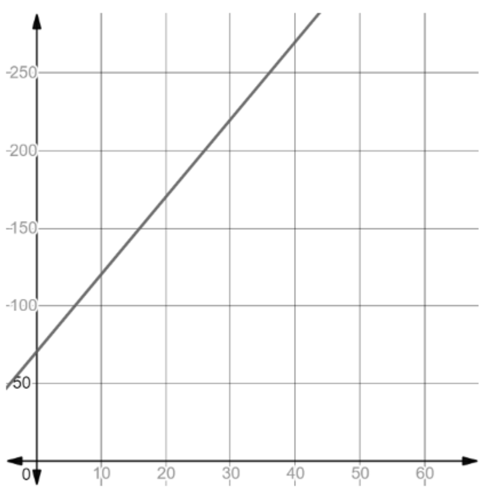
x	y
0	225
1	247
2	271
3	297
4	326

Given a graph, identify the function as linear, quadratic, or exponential. Explain your choice.

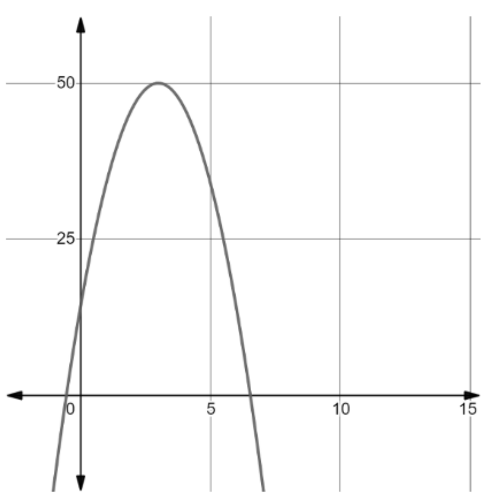
8.



9.



10.



Algebra 1 Name____________________
Unit 13 Date____________________
Lesson 11 Practice Period____________________

The table shows the number of tons of blueberries that were imported to the US from 2005 to 2013.

Year	Blueberries Imported (tons)
2005	26,319
2006	32,602
2007	35,125
2008	52,111
2009	60,450
2010	76,743
2011	87,336
2012	96,367
2013	103,502

1. Determine the average rate of change from 2010 to 2013.

2. Interpret the average rate of change for the interval 2010 to 2013 in terms of the context.

3. Determine the average rate of change from 2005 to 2009.

4. Interpret the average rate of change for the interval from 2005 to 2009 in terms of the context.

5. How does the average rate of change for the interval from 2005 to 2009 compare to the average rate of change for the interval 2010 to 2013?

The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, U.S. passengers traveled on a train for the period of 1970 to 2000, where x is the number of years since 1970.

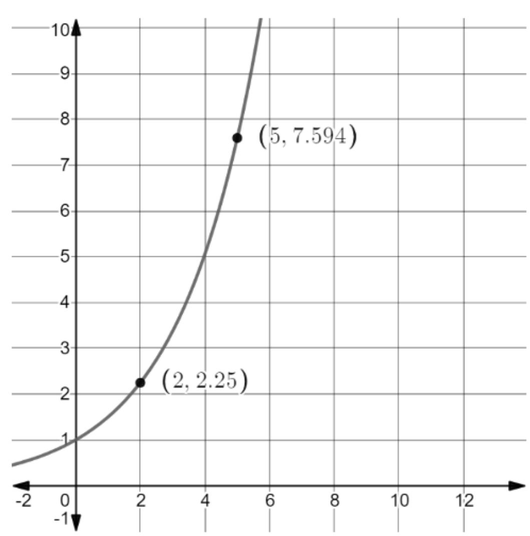
6. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1980 to 1985?

7. Interpret the average rate of change for the interval 1980 to 1985 in terms of the context.

8. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1985 to 1990?

9. Interpret the average rate of change for the interval 1985 to 1990 in terms of the context.

10. The graph of a function is shown. Points $(2, 2.25)$ and $(5, 7.594)$ are on the function. Calculate the average rate of change for the line that passes through the points $(2, 2.25)$ and $(5, 7.594)$.



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Lesson H2 Practice Period____________________

Use the functions $b(x) = -2x^2 + 6x - 3$ and $c(x) = 8 + 3x$ to determine each new function. Write your answer in standard form.

1. $(b + c)(x)$

2. $(b - c)(x)$

3. $(c - b)(x)$

4. $(b \cdot c)(x)$

5. $\left(\frac{b}{c}\right)(x)$

6. Write the restriction on the domain for $\left(\frac{b}{c}\right)(x)$.

Paulina and Stefan are renting a boat for a trip to the Keys. The boat costs $\$200$ per day. The rental includes 125 miles for free and then charges $\$0.50$ per mile. The function $B(m) = 0.50(m - 125) + 200$ models the cost of the boat, B, in terms of miles traveled, m. They determine gas will cost $\$1.75$ for each mile they travel. The company charges a $\$75$ fee for the first tank of gas. The function $F(m) = 1.75m + 75$ models the cost of gasoline for their vacation.

7. What is a reasonable domain for $B(m)$?

8. What is a reasonable domain for $F(m)$?

9. Determine the function that gives the total cost, $C(m)$, in dollars, based on the miles traveled.

10. What is a reasonable domain for $C(m)$?